Quantitative Research Memo: Path-Dependent Risk Analysis of Stochastic Bankroll Trajectories

Rafael Cristiano Nathanael Barus, November 2026

Executive Summary & Objective

In professional trading and high-stakes cash-game poker, looking at average returns is not enough. A player can maintain a positive long-term win rate and still go completely bankrupt if variance hits them at the wrong time. Now I'm not a trader, yet, but I do love some poker from time to time.

You hear poker pro's talk all about EV, but I've always thought, there's always a chance that you make positive EV plays all day and still end up down, right? You can make such good plays, but if you're really unlucky in the first few minutes/hands, you're limited in where you can go from there.

The phenomenon where the *sequence* of wins and losses dictates survival is known as **path-dependent risk**. Traditional static models assume smooth, predictable growth, ignoring the reality that a severe drawdown early on can trigger total capital depletion before a player ever reaches the long run.

To rigorously measure this risk, I built a **Monte Carlo simulation engine** in Python. Rather than treating bankroll growth as a straight line, the model simulates 500 parallel career paths over a 10,000-hand horizon, tracking how a starting bankroll reacts to real-world volatility, confidence intervals, and an **absorbing barrier at zero capital**. We'll start at 2000BB (20 standard buy-ins at 6-max No-Limit Hold'Em), as that's often the most recommended bankroll at any level.

2. Empirical Data & Parameterization

Every poker session leaves a paper trail, but turning raw logs into rigorous model parameters takes a bit of work. My tracking data is split across two distinct environments: high-volume online sessions and lower volume live (IRL) games. (In terms of time spent though, I've spent significantly much more playing in-person).

Medium	Total Hands Played (n)	Recorded Winnings (BB)	Per-Hand Expectation (μ)
Online Cash Games	7724	343.6	0.0445
Live (IRL) Cash Games	6876	2069	0.3105
Combined	**14600**	2412.6	**0.1698 (Weighted Average)**

Table 1. Poker Data (2025-2026)

From this, we have our first part of the equation, $\mu = 0.1698$

Because my tracking logs record summary totals rather than raw hand-by-hand histories, I couldn't compute an internal standard deviation directly from my own logs. Instead, I applied established empirical benchmarks for 6-max No-Limit Hold'em, which is $\sigma = 80{\sim}100$ per 100 hands, setting the hand-to-hand standard deviation to $\sigma = \frac{90}{\sqrt{100}} = 9$

To account for estimation error in my sample size, I computed 95% confidence intervals around my mean win rate. This gives us two distinct boundaries to stress-test against reality:

- **The Lower Bound (Running Cold):** $0.1698 - 1.96 * \frac{9}{\sqrt{14600}} = 0.023$ BB/hand
- **The Upper Bound (Running Hot):** $0.1698 + 1.96 * \frac{9}{\sqrt{14600}} = 0.315$ BB/hand

These parameters form the input matrix for the stochastic simulation engine, allowing us to test how different states of luck impact bankroll survival.

Something I personally noticed is that the lower bound is still positive, which is a shocker to me. However, is this number effective enough to protect me and my bankroll from variance, or when I'm in a particularly unlucky streak?

3. Stochastic Engine & Stress Testing

To evaluate how these parameters behave over a 10,000-hand career, I built a Monte Carlo simulation engine in Python (NumPy and Matplotlib).

Instead of treating bankroll progression as a deterministic straight line, the engine models **500** parallel career paths simultaneously. Each hand is drawn from a normal distribution defined by our win rate $0.023\ and\ 0.315$ and volatility $\sigma = 9$ tracking a cumulative path with an absorbing barrier at zero Big Blinds (representing total ruin). I ran the simulation twice under our 95% confidence interval boundaries to stress-test the strategy against extreme luck states.

```
Risk of Ruin: 1.00%
Percentage of careers ending down at 10k hands: 41.00%
--- FINAL BANKROLL SPREAD AT HAND 10,000 ---
Pessimistic (5th Pctl): 865.7 BB
Median (50th Pctl):     2204.6 BB
Optimistic (95th Pctl): 3593.4 BB
```

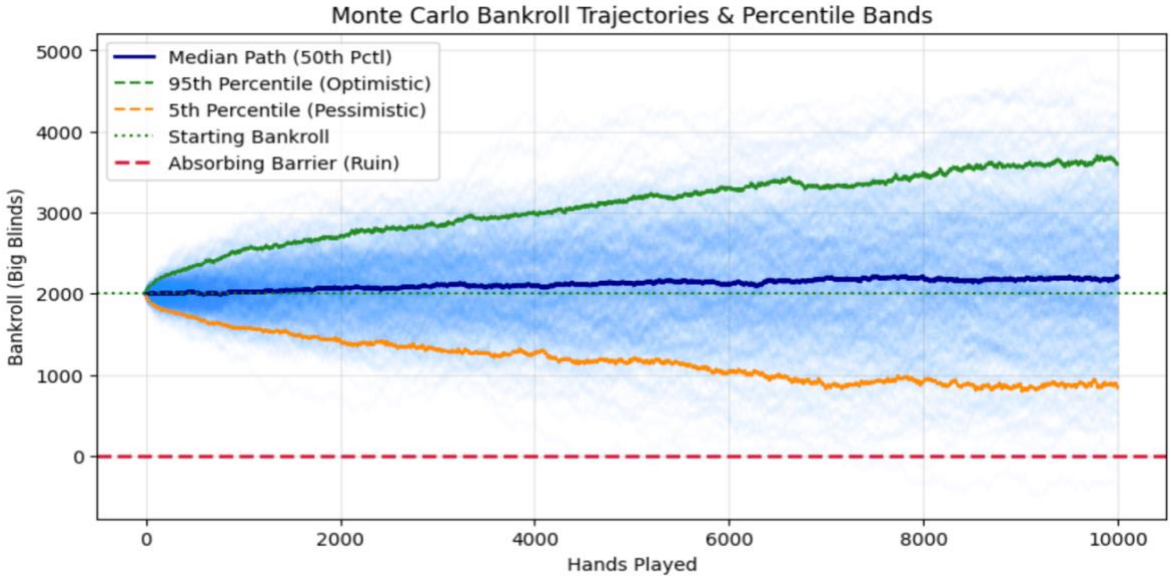


Figure 1. Monte Carlo simulation at lower bound win rate (0.023)

```
Risk of Ruin: 0.00%
Percentage of careers ending down at 10k hands: 0.00%
--- FINAL BANKROLL SPREAD AT HAND 10,000 ---
Pessimistic (5th Pctl): 3801.6 BB
Median (50th Pctl):     5078.0 BB
Optimistic (95th Pctl): 6537.1 BB
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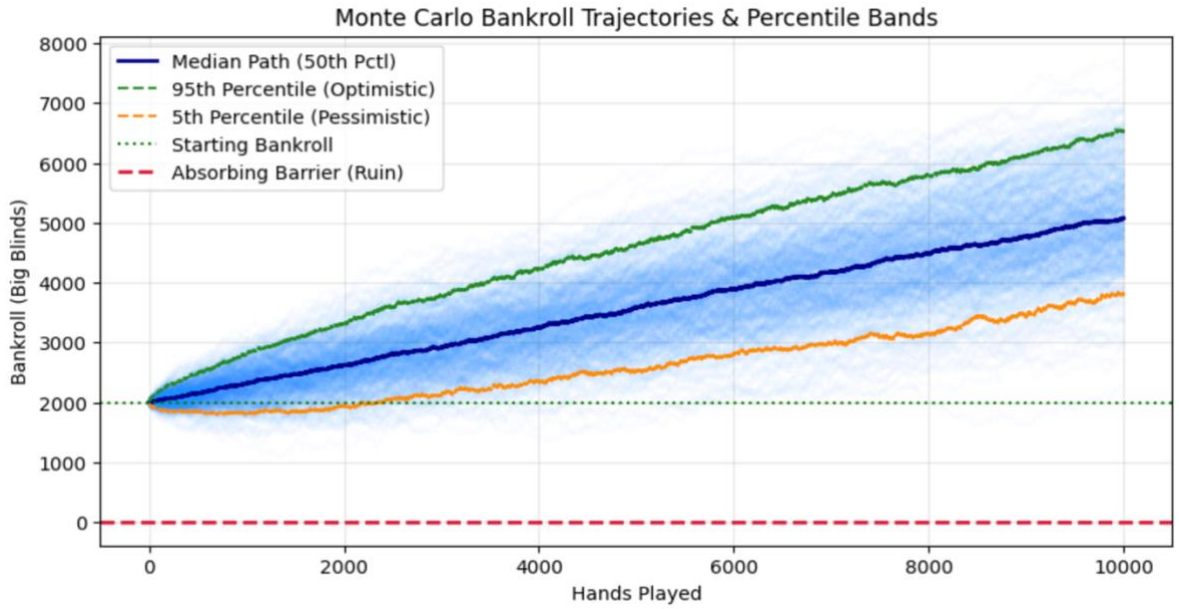


Figure 2. Monte Carlo simulation at upper bound win rate (0.315)

4. Comparison and Analysis

Firstly, I will justify why I'm doing it at a hand-by-hand basis instead of using the typical poker metrics of per 100 hands. While aggregating win rates in 100-hand blocks preserves expected value over large samples, it smooths out intra-block volatility. Simulating hand-by-hand ensures that capital depletion to zero (the absorbing barrier of ruin) is captured at the exact hand it occurs, preventing mid-block bankruptcies from being artificially masked by subsequent recovery within the same interval. Simply, you can't play till hand 200 if you bust out at hand 101.

When you look at the numbers, hitting absolute zero (the Risk of Ruin), isn't necessarily a problem for me. Even in the unluckiest of streaks (Running Cold lower bound), I'd only have

a 1% risk of ruin. However, what is more worrying is the final amount, or the terminal value after 10000 hands.

Under the "Running Cold" bound, a staggering 41% of simulated careers finish in the red, having been better off not playing the game at all. Even though the math says I technically have a positive win rate, variance of this sample size is still so brutal that I can end up worse than how I started. If we do the math, over 10000 hands my estimated win-rate should be $10000 * 0.023 = 230$ BB, however, the variance of this sample is also $9 * \sqrt{10000} = 900$BB. That leaves a massive variance swing of $-670\ to + 1130$BB.

When we compare that to the "Running Hot" bound, the 0% result highlights that I can be pretty certain that I will be profitable over 10000 hands.

To move beyond single-point averages, the simulation also evaluates terminal wealth distributions across a 10,000-hand horizon.

- **Terminal VaR (95% Confidence):** Across the distribution of simulated career paths, the **5th percentile (p5) terminal boundary** establishes that 95% of careers will finish *above* a bankroll of 865.7BB (1,134BB loss) running cold, and 3801BB running hot. Only 5% of simulated trajectories experience severe degradation or ruin below this threshold.
- **Upside Potential:** Conversely, the **95th percentile (p95) boundary** highlights that top-decile running outcomes scale past 3593BB running cold, and a staggering 6537BB running hot, capturing the asymmetric upside of a positive expectation stream compounded by variance.

5. Potential Drawbacks

The Gaussian Normal Assumption (Fat Tails): This Monte Carlo simulation follows a normal distribution, but are poker outcomes normally distributed? Obviously not. You're much more likely to stack someone's entire buy-in, instead of just 70-80% of it. You also can't win more than someone's entire stack of chips, obviously. The model assumes that hand-to-hand outcomes follow a clean, normal distribution. Poker and financial markets are

leptokurtic meaning they feature "fat tails." Extreme outliers, coolers, and multi-way set-over-sets happen more frequently in actual play than a standard bell curve predicts.

Stationary Win Rate: The simulation assumes a constant, fixed win rate across all 10,000 hands. In live play, your edge is non-stationary; it fluctuates based on table softness, opponent tilt, fatigue, game selection, and your own mental state.

Absorbing Barrier Realities: The model treats going down to 0BB as an instant, permanent end state. In practice, bankroll depletion leads to psychological tilt, forced movement down in stakes (moving down in limits), or reloading funds rather than absolute career termination.

6. Conclusion

At the end of the day, poker, much like quantitative trading, isn't just about having a positive expected value, but a survival test against variance.

As this simulation shows, a 2000 BB bankroll is mostly safe from absolute bankruptcy under normal conditions, but surviving isn't the same as thriving. With variance from the game, or from the real world, some iterations are better off not playing at all.

By treating bankrolls as a stochastic path rather than a static average, we can map out the real dangers of the grind. Personally, it changed how I viewed a downswing: no longer as a failure of skill, but as an inevitable mathematical hurdle that requires proper capital preservation to weather.

Sure, you can make the correct move every single time; and in the long run, yeah, you'll do well. We've found that survival isn't a big issue, but your edge still has to be big enough to protect yourself from variance, or else you might as well shouldn't start at all.

Link to the simulation code: https://github.com/RafaCristiano7/Poker-Monte-Carlo-Bankroll-Simulation